

CS180 Project 2

Fun with Filters and
Frequencies

EXPORT

SETTINGS

Fit • 100%

DEVELOP

GALLERY

COLLECTIONS

All Photos 7

Recently Added

Favorites

PART 1: FUN WITH
FILTERSConvolutions
from Scratch!Finite Difference
OperatorDerivative of
Gaussian (DoG)PART 2: FUN WITH
FREQUENCIES!Image
"Sharpening"

Hybrid Images

Gaussian and
Laplacian StacksMultiresolution
Blending

QUICK ACCESS

Deleted 0

Part 1.1: Convolutions from Scratch!

Let's implement convolutions with four for loops, then two for loops. I'll compare it with `scipy.signal.convolve2d`!

Implementation

```
# Two-loop convolution implementation
def convolve2l(im1, filt):
    if filt.ndim == 1:
        filt = filt.reshape(1, -1) if len(filt) > 1 else filt.reshape(1, 1)

    filt = np.flip(filt, axis=(0,1))
    cx = filt.shape[0] // 2
    cy = filt.shape[1] // 2
    convolved = np.zeros(im1.shape)
    padded = np.zeros((im1.shape[0] + 2*cx, im1.shape[1] + 2*cy), dtype=float)

    if cx == 0 and cy == 0:
        padded = im1
    elif cx == 0:
        padded[:, cy:-cy] = im1
    elif cy == 0:
        padded[cx:-cx, :] = im1
    else:
        padded[cx:-cx, cy:-cy] = im1

    # Convolve entire image
    for r in range(padded.shape[0] - (filt.shape[0]-1)):
        for c in range(padded.shape[1] - (filt.shape[1]-1)):
            convolved[r,c] = np.sum(np.multiply(filt,
                padded[r:r+filt.shape[0], c:c+filt.shape[1]]))

    return convolved

# Four-loop convolution implementation
def convolve4l(im1, filt):
    # Preprocessing (padding) implementation not shown

    # Convolve entire image with four nested loops
    for r in range(padded.shape[0] - (filt.shape[0]-1)):
        for c in range(padded.shape[1] - (filt.shape[1]-1)):
            filtered = 0
            for i in range(filt.shape[0]):
                for j in range(filt.shape[1]):
                    filtered += filt[i,j] * padded[r + i, j + c]
            convolved[r,c] = filtered
    return convolved
```

Results

Here is my selfie and a convolution with my implementation

